Literature Review

Project: LinkedIn Job Market Analysis — What Drives Demand and Pay for Data Roles
Track: Junior Data Analysis — DEPI **Version / Date:** 1.0 — 15 Oct 2026

1. Purpose

This review places the project in the context of existing work on (a) job-posting data as a source of labour-market evidence, (b) skill demand and pay, (c) data modelling and DAX for analytics, and (d) dashboard design. It also compares existing tools and identifies the gap this project fills.

2. Job-Posting Data as Labour-Market Evidence

Online vacancy data has become a common source for studying labour demand because it is large, timely and detailed compared with traditional surveys. Hershbein and Kahn (2018) used large-scale vacancy postings to study how employers change skill requirements over the business cycle, showing that posting text and requirements carry real economic signal. Deming and Kahn (2018) used job-posting data to link skill requirements to firm performance and wages across markets. These studies support the project's premise that posting-level fields such as title, experience level, skills and salary can be analysed to explain demand and pay.

Known limitations in this literature (and relevant here): postings do not equal hires, they over-represent some industries and larger employers, and salary is only disclosed in a share of postings. In our dataset salary is available for about 29% of postings, so salary findings must be reported with their coverage.

3. Skills, Demand and Pay for Data Roles

The growth of data roles is widely documented. Davenport and Patil (2012) described the data scientist as one of the most in-demand professions, and the World Economic Forum's *Future of Jobs Report* (2023) lists data analysts and data specialists among the fastest-growing roles, with analytical thinking and technology literacy among the most important skills. Together they motivate a focus on analyst roles and on which skills are rewarded with higher pay.

The project applies the same logic at posting level: it counts how often each of the 35 standardised skill categories appears in postings and relates them to salary, experience level and industry.

4. Data Modelling and DAX

For analytical reporting, the dimensional (star schema) approach by Kimball and Ross (2013) is the standard reference: a central fact table surrounded by dimension tables, which simplifies filtering and improves performance. Here, `postings` is the fact table, while companies, skills, industries and benefits are dimensions or bridge tables. Because jobs can have several skills and industries, bridge tables and distinct counts are required to avoid double counting.

Ferrari and Russo (2019) explain how DAX measures depend on filter context and relationships, which is the basis for measures such as distinct job counts, median salary, apply rate and percent-of-total.

5. Dashboard Design

Few (2013) recommends dashboards that fit on one screen, put the most important information first, and use simple visual encodings. Knaflic (2015) stresses clear titles, removing clutter and guiding the viewer to the message. These principles are used to design the six dashboard pages, with KPI cards at the top, consistent colours, and one message per page.

6. Existing Tools and Solutions

Tool	What it offers	Limitation for this project
LinkedIn Talent Insights / LinkedIn Jobs	Talent and hiring analytics for companies	Paid, closed, no access to raw data
Glassdoor / Indeed Hiring Lab	Salary and hiring trend reports	Aggregated reports, not reproducible
Kaggle notebooks on the same dataset	Mostly exploratory charts in Python	Few full, interactive dashboards; little on data modelling or DAX
Excel pivot analysis	Quick summaries	Cannot handle about 500 MB / 123,849 postings with 10 linked tables comfortably
Tableau vs Power BI	Both support interactive dashboards	Power BI chosen: DAX, strong modelling, taught in the DEPI track

7. Gap and Contribution

Existing public work on this dataset is largely exploratory. This project contributes:

1. A **documented star-schema model** that joins all 10 source tables correctly.
2. A library of **DAX measures** handling many-to-many skill and industry links.
3. A **six-page interactive dashboard** with a dedicated Data/Business Analyst deep-dive.
4. **Transparent data-quality reporting** (coverage of salary, remote flag and applications).

8. Conclusion

The literature supports using job-posting data to study demand, skills and pay, and supports dimensional modelling plus clear dashboard design as the way to communicate it. The main cautions are limited salary coverage, a short time window and selection bias, which are carried into the project's requirements and risk plan.

References

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5. Hershbein, B., & Kahn, L. B. (2018). Do Recessions Accelerate Routine-Biased Technological Change? Evidence from Vacancy Postings. *American Economic Review*, 108(7).
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7. Knaflic, C. N. (2015). *Storytelling with Data*. Wiley.
8. World Economic Forum (2023). *The Future of Jobs Report 2023*.
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Please verify each reference's details before submission.